

Leveraging Error-Tolerance Asymmetry in Automated Electrical Grid Visual Inspection with a Semi-Supervised Annotation Pipeline

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Background

- Improving **Electrical Grid Reliability** requires **Efficient Maintenance Operations**
- UAV Data Collection** enables efficient scaling of **Visual Inspection Operations**
- Human Inspectors** focus on defective components, producing **Sparse Annotations**
- Precise Geometric Annotations** have no direct benefit for **Inspection Reporting**

Challenges

- Developing **High-capacity, Task-specific Models** requires **Large-scale, Annotated Datasets**
- Inspection Reporting** and **Supervised Learning** have different **Data Labeling Requirements**
- Processing **High-resolution** imagery is **Impractical**, while **Downsampling** risks **Losing critical Details**.
- Spatially Imprecise Annotations** can introduce **Misleading Training Signals**

Approach

- Leverage Dataset Construction** with **Model-assisted** labeling for **Non-defective** instances
- Account for **Error-Tolerance Asymmetry** by reserving **Labeling Effort** for **Critical** instances
- Process Relevant ROIs** by **Cropping** raw images based on **Instance Locations**
- Refine Spatial Supervision** through **Visual Prompting** of foundation models

Data

Instances

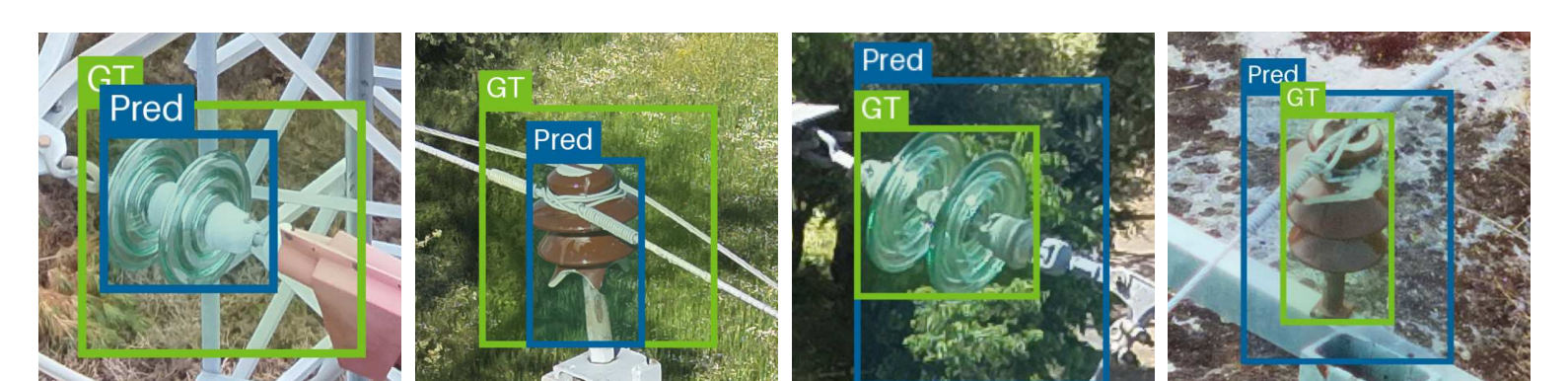
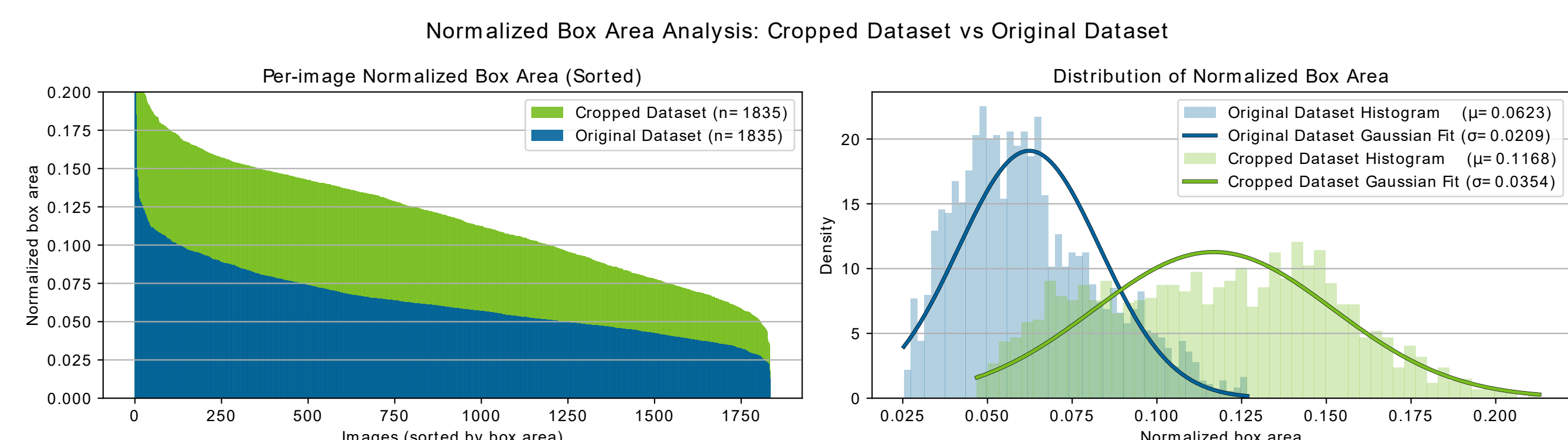
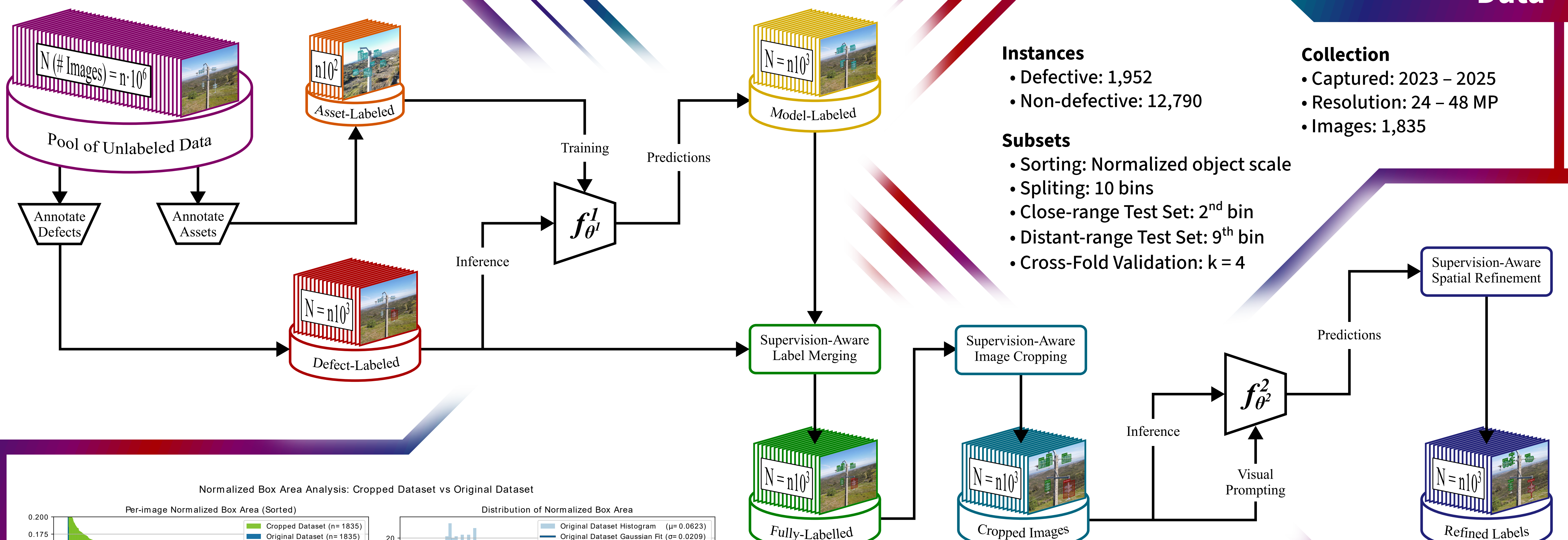
- Defective: 1,952
- Non-defective: 12,790

Subsets

- Sorting: Normalized object scale
- Splitting: 10 bins
- Close-range Test Set: 2nd bin
- Distant-range Test Set: 9th bin
- Cross-Fold Validation: $k = 4$

Collection

- Captured: 2023 – 2025
- Resolution: 24 – 48 MP
- Images: 1,835



IoU	0.388783	0.390583	0.404719	0.415551
IoR	1.000000	1.000000	0.404719	0.415551
IoP	0.388783	0.390583	0.999996	1.000000

Results

AP@IoU≥50 — Dataset Improvements

- Close-range: $\Delta\mu +5.0\% \cdot \Delta\sigma -1.8\%$
 $0.886 \pm 0.031 \rightarrow 0.936 \pm 0.013$
- Distant-range: $\Delta\mu +16.0\% \cdot \Delta\sigma -1.6\%$
 $0.569 \pm 0.041 \rightarrow 0.729 \pm 0.025$

AP@IoR≥50 — Metric Selection

- Full Test Set: $\Delta\mu +1.0\% \cdot \Delta\sigma -0.8\%$
 $0.778 \pm 0.138 \rightarrow 0.788 \pm 0.130$

- Contextual Supervision** improves **Training Convergence**: Adding non-defective asset labels increases mean performance and reduces variance

- ROI Cropping** benefits **Distant Images**: Small object scale limits signal quality, highlighting the importance of matching receptive fields to cue size

- Spatial Detail** improves **Gradient Quality**: Polygonal refinement provides consistent additional gains

- Ground-Truth Noise** can **Hide Performance**: IoR and IoP reduce sensitivity to spatial annotation imprecision, revealing detections missed by strict IoU

Findings

